

Sarang K

Kannur, Kerala | +91 8078738602 | sarangsajeevan629@gmail.com
linkedin.com/in/sarangk2003 | github.com/Sarangk2024 | Portfolio

Professional Summary

Backend-focused Computer Science engineer with hands-on experience designing and deploying RESTful APIs, machine learning inference pipelines, and containerized applications using Python, FastAPI, and Node.js. Delivered working software across two internships and three independently scoped projects spanning AI, blockchain, and IoT domains. Comfortable working across the full backend lifecycle — from schema design and API development to model training and cloud deployment. Targeting entry-level Software Developer roles where backend engineering and scalable system design create measurable impact.

Technical Skills

Languages: Python, JavaScript, Java, C++

Frontend: React, HTML, CSS, JavaScript (ES6+)

Backend & APIs: FastAPI, Node.js, Express.js, ASP.NET MVC, RESTful API Design, JWT

Databases: SQL, PostgreSQL, Query Optimization

DevOps & Cloud: Docker, Git, CI/CD Pipelines, Render, Vercel

AI / ML: EfficientNet, MobileNetV2, TensorFlow, Image Classification, Model Training & Evaluation

Engineering Practices: Agile/Scrum, OOP, DSA, DBMS, Debugging, Postman, Swagger

Experience

AI Intern | Infosys Springboard

Dec 2025 – Mar 2026

- Trained an EfficientNet transfer learning model for plant disease classification across **10 crop categories**, improving validation accuracy from **80% to 92%** through iterative preprocessing, augmentation, and hyperparameter tuning
- Designed and deployed the complete ML pipeline — data ingestion, augmentation, model training, and inference — using TensorFlow and Python; validated performance on held-out test sets and documented results for reproducibility

Software Developer Intern | Eqsoft Business Solutions Pvt Ltd

Jun 2025 – Jul 2025

- Developed **12+** REST API endpoints using ASP.NET MVC to power client-facing business workflows, backed by a normalized MySQL database
- Implemented JWT-based authentication and 3-tier Role-Based Access Control (RBAC); optimized MySQL queries through index tuning, reducing average execution time by **35%**

Projects

AgriDetect AI | Python, FastAPI, EfficientNet, Docker

Live Demo

- Developed a publicly deployed ML web application for crop disease detection across **10 categories** (wheat, apple, grape, rice, pumpkin, and others), achieving **92% accuracy** using EfficientNet with transfer learning
- Architected **8** FastAPI endpoints with JWT-secured access and sub-300ms average response time; containerized with Docker and deployed via automated CI/CD pipeline to Vercel

Blockchain Product Verification System | Node.js, Express.js, MongoDB, React

Live Demo

- Engineered **10** backend API endpoints for product authentication and blockchain verification using Node.js and Express.js, with full Postman-tested documentation
- Integrated Web3.js for on-chain transaction calls on **Ethereum testnet (Sepolia)** and designed an indexed MongoDB schema to prevent duplicate and fraudulent product registration

Smart Helmet Accident Alert System | PHP, MySQL, IoT

- Engineered a PHP backend to ingest real-time accelerometer and GPS data from helmet-mounted IoT sensors, triggering automated emergency contact notifications within **500ms** of collision detection
- Designed a normalized MySQL schema across **5 tables** to store sensor readings, user profiles, and alert event logs, enforcing referential integrity across all foreign key relationships

Education

B.Tech in Computer Science and Engineering

2022 - 2026

College of Engineering Thalassery, Kerala

CGPA: 8.88 / 10

Certifications

Infosys AI Virtual Internship Certificate, IBM Full Stack Software Developer Professional Certificate, UPCODE AI Bootcamp Certificate